PAPER SEMINAR

Beyond Accuracy: Behavioral Testing of NLP Models with CheckList

Ribeiro, Wu, Guestrin & Singh · ACL 2020

Leonardo · Eduarda · Luis · Valeria

AI-generated content (Claude). Review before use.

PART 1 · MOTIVATION

The problem: accuracy hides failures

Biased test sets

Same biases as the training data

One number

Does not show where the model fails

Narrow alternatives

One task or one capability at a time

BERT: **92.7%** on SST-2 — yet fails 85% of “I thought the plane would be awful, but it wasn’t.”

Ribeiro et al. (2020), §1 and Table 1

PART 1 · CONTRIBUTIONS

What the paper contributes

1

A method

Test NLP models like software: one skill at a time

2

A free tool

Generates thousands of tests automatically

3

Evidence

Bugs in BERT and in Microsoft, Google, Amazon; users find 3× more bugs

Ribeiro et al. (2020), §1 and §6 · github.com/marcotcr/checklist

PART 1 · RELATED WORK

Before CheckList: one piece at a time

Accuracy

Only one number

e.g. SQuAD leaderboard (Rajpurkar et al., 2016)

Robustness and fairness tests

One skill or one task only

Belinkov & Bisk, 2018 · Prabhakaran et al., 2019

Challenge sets

Small and hand-made

Naik et al., 2018

Knowledge tests

The model knows it, but doesn’t use it

Tenney et al., 2019

CheckList brings these together, using ideas from **software testing**.

Ribeiro et al. (2020), §1 and §5

From accuracy to behavioral testing

Held-out accuracy

% correct on unseen examples
One number for everything

Behavioral testing

Black box: inputs → outputs
One behavior per test

Like a school report card per subject, not just the final grade.

Ribeiro et al. (2020), §1; Beizer (1995)

PART 2 · KEY CONCEPTS

Capabilities, perturbations, metamorphic tests

Capability

A skill the task needs

Negation: “not poor” ≠ negative

Perturbation

A controlled input change

Chicago → Dallas

Metamorphic test

Compare outputs, no answer key

Same city swap → same sentiment

Ribeiro et al. (2020), §2; Segura et al. (2016)

The tasks and models

Task	The model must…
Sentiment analysis	label text positive, negative or neutral
Quora Question Pairs	decide if two questions are duplicates
Reading comprehension	answer a question from a paragraph (SQuAD)

BERT / RoBERTa: pretrained Transformer language models, fine-tuned per task. A **masked LM** fills in a hidden word: “I really [mask] the flight.”

Ribeiro et al. (2020), §1 and §3

A matrix of capabilities × test types

Capability	MFT	INV	DIR
Vocabulary	15.0%	16.2%	34.6%
Named entities	0.0%	20.8%	—
Negation	**76.4%**	—	—

Each cell = failure rate of a commercial model. Black box: no access to internals needed.

Ribeiro et al. (2020), Figure 1

Three test types

MFT

Simple labelled cases

“I didn’t love the flight.” → negative

INV

Change should not matter

Cuba → Canada: same prediction

DIR

Change has a known effect

+ “You are lame.”: not more positive

INV and DIR need **no labels**.

Ribeiro et al. (2020), §2.2

PART 3 · METHODOLOGY

Generating tests and experimental design

I {NEGATION} {POS_VERB} the {THING}.

didn’t / can’t say I · love / like · food / flight / service → every combination is a test · plus RoBERTa word suggestions, name lists, typos

A · Test the models

3 tasks · Microsoft, Google, Amazon APIs + BERT, RoBERTa · failure rate per test

B · Test with users

Microsoft team, 5 h · 18 users, 2 h, 3 groups of 6

Ribeiro et al. (2020), §2.3, §3 and §4

PART 4 · PRIMARY RESULTS

“Solved” models fail the basics; users find more bugs

Models (failure rate)

100% “…awful, but it wasn’t.” (Microsoft, Amazon)

100% he / she coreference (reading comprehension)

98% “before” vs. “after” (BERT, QQP)

89% “John is not a doctor, Mary is.” → John

Users (full tool vs. none)

2× tests created (13.5 vs. 5.8)

27× cases per test (198 vs. 7.3)

≈3× serious bugs (6.2 vs. 2.2)

Microsoft: new bugs in a mature model

Ribeiro et al. (2020), Tables 1–4 and §4.1 · n = 6 users per group

PART 4 · APPLICATIONS

Main applications of CheckList

Choosing a model

Compare models skill by skill, not by one score — e.g. research vs. commercial

Development cycle

Re-run the same suite on every new model version, as Microsoft’s team planned

Benchmarks and pipelines

Add the tests to existing benchmarks and evaluation pipelines

Shared test suites

Teams build and reuse tests together; many transfer across tasks

Works on any model, even a paid API: no access to data or weights needed.

Ribeiro et al. (2020), §3, §4.1 and §6

PART 4 · CONCLUSION

Accuracy is useful, but not enough

Takeaways

- Works for any task and model
- Finds bugs even in products
- Easy to learn

Limitations

- Behavior only (not labels, security)
- Passing ≠ correct
- Small user study

From *“how accurate is it?”* to *“what can it do?”*

Ribeiro et al. (2020), §5–6 · “small user study” is the presenters’ observation

References

- **Ribeiro, M. T., Wu, T., Guestrin, C., & Singh, S. (2020).** Beyond accuracy: Behavioral testing of NLP models with CheckList. *Proc. ACL*, 4902–4912.
- Beizer, B. (1995). *Black-box testing.* Wiley.
- Belinkov, Y., & Bisk, Y. (2018). Synthetic and natural noise both break neural machine translation. *Proc. ICLR*.
- Naik, A., Ravichander, A., Sadeh, N., Rose, C., & Neubig, G. (2018). Stress test evaluation for natural language inference. *Proc. COLING*.
- Prabhakaran, V., Hutchinson, B., & Mitchell, M. (2019). Perturbation sensitivity analysis to detect unintended model biases. *Proc. EMNLP-IJCNLP*, 5740–5745.
- Rajpurkar, P., Zhang, J., Lopyrev, K., & Liang, P. (2016). SQuAD: 100,000+ questions for machine comprehension of text. *Proc. EMNLP*, 2383–2392.
- Segura, S., et al. (2016). A survey on metamorphic testing. *IEEE TSE*, 42(9), 805–824.
- Tenney, I., Das, D., & Pavlick, E. (2019). BERT rediscovers the classical NLP pipeline. *Proc. ACL*.

Tool: github.com/marcotcr/checklist

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